Week 3

Practice Problem

Question 6:

$f: \mathbb{R} \to \mathbb{R}$.

Let f be cts.

Let $U \subseteq \mathbb{R}$ be open set.

claim $f^{-1}(U) = \{x \in \mathbb{R} \mid f(x) \in U\}$ is open.

Let

$x_0 \in f^{-1}(U) \Rightarrow f(x_0) \in U$

Since U is open

$\Rightarrow \exists\ \mu > 0$ s.t. $(f(x_0) - \mu, f(x_0) + \mu) \subseteq U$

Now,

for $\epsilon > 0$, $\mu > \epsilon > 0$,

we get $\delta > 0$ s.t. $|x - x_0| < \delta$

we have

$|f(x) - f(x_0)| < \epsilon$

i.e. $\Rightarrow$

as $x \in (x_0 - \delta, x_0 + \delta)$

$\Rightarrow f(x) \in (f(x_0) - \epsilon, f(x_0) + \epsilon)$

$\Rightarrow (x_0 - \delta, x_0 + \delta) \subseteq f^{-1}(U)$

$\Rightarrow$ Since x_0 is arbitary in U so $f^{-1}(U)$ is open.

— Conversely:

$\Leftarrow$ let if $U \subseteq \mathbb{R}$ open then $f^{-1}(U) = \{x \in \mathbb{R} : f(x) \in U\}$ is open

claim; f is continuous.

let, $x_0 \in \mathbb{R}$ be any arbitary number. let $f(x_0) \subseteq U$ (containing in some open set)

$\Rightarrow$ for $\epsilon > 0$, $(f(x_0) - \epsilon, f(x_0) + \epsilon) \subseteq U$

Now, ~~for $x \in (f(x_0) - \epsilon, f(x_0) + \epsilon)$~~

~~$f^{-1}(x) \in f(U)$~~

and $f^{-1}(U)$ is open $\Rightarrow \exists\, \delta > 0$
so $(x_0 - \delta, x_0 + \delta) \subset U$
Thus,
for ~~$x \in (x_0 - \delta, x_0 + \delta)$~~
$\epsilon > 0$ $\exists\, \delta > 0$
s.t. $|f(x) - f(x_0)| < \epsilon$
whenever $|x - x_0| < \delta$.

(b) Similarly proof, for instead interval just take ball around the point x_0.

Ans. 8. $E \in \mathcal{B}_{\mathbb{R}}$
= A "open in E" if $\exists\, B \subset \mathbb{R}$
s.t. $A = B \cap E$

- Now: Claim
$\mathcal{B}_E : \sigma(\text{all open set in } E) = \mathcal{B}_{\mathbb{R}} \cap E$. —
Let
- $A \subseteq E$ be open set in E
$\Rightarrow$ $A = B \cap E$ for some $B \in \mathcal{B}_{\mathbb{R}}$
$\Rightarrow$ $\{\text{all open set in } E\} \subseteq \mathcal{B}_{\mathbb{R}}$ —— (1)
- Claim:- $\mathcal{B}_{\mathbb{R}} \cap E$ is σ-field.
Let ~~$\mathbb{R} = \mathbb{R} \setminus A$~~
□ $E = E \cap \mathbb{R}$
□ Let $A \in \mathcal{B}_{\mathbb{R}} \cap E$ $\Rightarrow$ $A = B \cap E$ for some E
Now $A^c = B^c \cap E$ if $B \in \mathcal{B}_{\mathbb{R}} \Rightarrow B^c \in \mathcal{B}_{\mathbb{R}}$
□ Let $A_1, A_2, \ldots A_n \in \mathcal{B}_{\mathbb{R}} \cap E$
$\Rightarrow$ $A_i = B_i \cap E$
So $\bigcup_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^{\infty} B_i\right) \cap E \in \mathcal{B}_{\mathbb{R}} \cap E \quad \because \left(\bigcup_{i=1}^{\infty} B_i\right) \in \mathcal{B}_{\mathbb{R}}$ —

So $\mathcal{B}_{\mathbb{R}} \cap E$ is σ-field. — (2)

So from (1) & (2)

$\mathcal{B}_E := \sigma$(all open set in E) $\subseteq \mathcal{B}_{\mathbb{R}} \cap E$ — (3)

Now, for all B open in $\mathcal{B}_{\mathbb{R}}$

$B \cap E \in \mathcal{B}_E$

$\because$ $\mathcal{B}_E$ is a σ-field

$\Rightarrow$ $\sigma(B \cap E) \subseteq \mathcal{B}_{\mathbb{R}}$.

and

$\sigma(B \cap E) = \mathcal{B}_{\mathbb{R}}$

$\Rightarrow$ $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{B}_E$ — (4)

from (3) & (4) $\mathcal{B}_E = \mathcal{B}_{\mathbb{R}} \cap E$.

Ques 9: $X : (\Omega, \mathcal{F}, P) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ be a R.V.

Let

$\sigma(X) := \{ X^{-1}(A) : A \in \mathcal{B}_{\mathbb{R}} \}$

claim $\sigma(X)$ is a σ-field.

Let $\mathbb{R} \in \mathcal{B}_{\mathbb{R}}$

then $X^{-1}(\mathbb{R}) = \Omega \in \sigma(X)$.

- Let $B \in \sigma(X)$

$\Rightarrow$ $X^{-1}(A) = B$ for some $A \in \mathcal{B}_{\mathbb{R}}$

Now, $B^c = (X^{-1}(A))^c$

$= (X^{-1}(A^c))$

$\Rightarrow \exists\, A^c \in \mathcal{B}_{\mathbb{R}}$ s.t. $X^{-1}(A^c) = B^c$

So $B^c \in \sigma(X)$

- Let $B_1, B_2 \ldots B_n \ldots \in \sigma(X)$

$\Rightarrow \exists\, A_1, A_2 \ldots A_n$ s.t. $X^{-1}(A_i) = B_i$

Now,

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} X^{-1}(A_n) = X^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right)$$

Since $\mathcal{B}_{\mathbb{R}}$ is σ-field

$\Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{B}_{\mathbb{R}} \Rightarrow \bigcup_{n=1}^{\infty} B_n \in \sigma(X)$.

Hence $\sigma(X)$ is σ-field.

Qu 10: $f: \Omega \longrightarrow (\Omega', \mathcal{F})$

$f: \{A, B, C\} \longrightarrow (\{1,2,3\}, \{\Omega', \phi, \{1\}, \{2,3\}\})$

Say:

$f(A) = 1$

$f(B) = 2$

$f(C) = 3$

$\sigma(f^{-1}(\mathcal{F})) = \sigma\{\{A \in \Omega / f(A) \in \mathcal{F}\}\}$

$= \sigma\{\Omega, \phi, A_1\}$

$= \{\Omega, \phi, A_1, A_1^c\}$.

Now:

$f^{-1}(\sigma(\mathcal{F})) = f^{-1}(\mathcal{F}) = f^{-1}(\{\Omega', \phi, \{1\}, \{2,3\}\})$

$= \{\Omega, \phi$

$= \{f^{-1}(\Omega'), f^{-1}(\phi), f^{-1}(\{1\}), f^{-1}(\{2,3\})\}$

$f^{-1}(\Omega) = \{\omega : \omega \in \Omega / f(\omega) \in \Omega'\}$

$= \{\Omega, \phi, A_1, A_1^c\}$.

General proof: claim $\sigma(f^{-1}(\mathcal{F})) = f^{-1}(\sigma(\mathcal{F}))$

let

$x \in \sigma(f^{-1}(\mathcal{F}))$

$\Rightarrow$

$x \in \sigma(\{\omega \in \Omega / f(\omega) \in A, A \in \mathcal{F}\})$

$\Rightarrow x \in$